

## CMSC 313 Homework 2

Name: \_\_\_\_\_

Assigned: 9/8/2026

Due: 9/16/2026 at 11:59pm

Total Points: 100

### Homework Setup

In this assignment, you will learn how to design and simulate digital circuits using the tool Logisim Evolution. Navigate to this GitHub page and select the download for Logisim Evolution v4.0.0 corresponding to your computer's operating system and architecture. Then, install it.

- <https://github.com/logisim-evolution/logisim-evolution/releases>

If you are unsure of which download to select, refer to the instructions here:

- <https://github.com/logisim-evolution/logisim-evolution?tab=readme-ov-file#download>

Save your assignment as HW2\_LASTNAME.circ, replacing LASTNAME with your last name in all capital letters. Do **NOT** rename any of the input/output pins. Do **NOT** change the vertical order of any input pins. Do **NOT** change the vertical order of any output pins. You are allowed to add labels to the logic gates in your circuit, but you are not required to. **Failing to follow these instructions will be a 50 point deduction from this assignment's grade.**

### Part 1: XOR Circuit Diagram (30 pts)

Switch to the XOR\_tutorial circuit by double-clicking the text "XOR\_tutorial" on the explorer pane. (You should switch to a blank view with the "A", "B", and "Q" pins, and you should see a red magnifying glass icon next to XOR\_Tutorial). Select "Help -> Tutorial" from the top navigation pane. Then navigate to "Guide to Being a Logisim User" -> "Beginner's Tutorial". Read through the Beginner's Tutorial page, then read and complete Steps 0 through 4 of the Logisim tutorial. **Make sure to complete these steps inside of the XOR tutorial subcircuit on the explorer pane, not the main subcircuit!** (the XOR\_tutorial entry on the explorer pane should have a red magnifying glass on it)

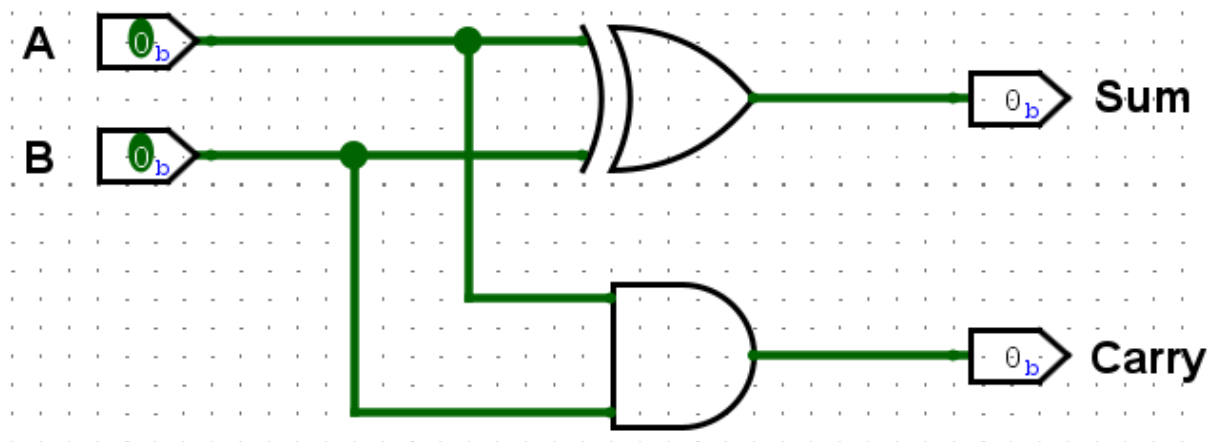
- Step 0: Orienting Yourself
- Step 1: Adding gates
- Step 2: Adding wires
- Step 3: Adding text

- Step 4: Testing your circuit

## **Part 2: Half Adder (15 pts)**

Save your progress, then switch to the Half\_adder circuit. A half-adder performs addition between two bits, A and B. It outputs the result in its sum output. If the sum would be a two-bit number, the carry output is set to 1.

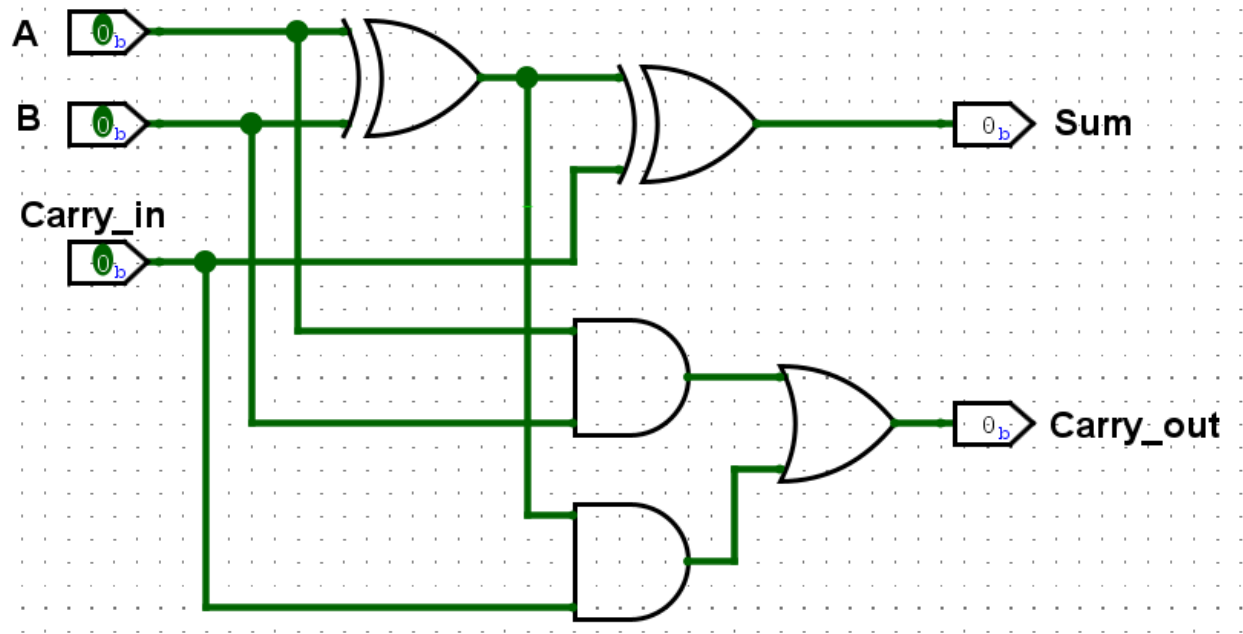
Re-create the circuit below in Logisim. However, instead of the default XOR gate from Logisim, use the XOR\_tutorial logic gate that you created in Part 1. **You will not receive credit if you do not use the XOR tutorial gate instead of the default XOR gate.**



## **Part 3: Full Adder (25 pts)**

Save your progress, then switch to the Full\_adder circuit. A full adder is similar to a half adder, but it takes a third (carry) input. The carry input is 1 if the carry output from a previous addition (performed by a different half adder or full adder) was 1.

Re-create the circuit below in Logisim. Instead of the default XOR gate from Logisim, use the XOR\_tutorial logic gate that you created in Part 1. **You will not receive credit if you do not use the XOR tutorial gate instead of the default XOR gate.**



#### **Part 4: main (30 pts)**

Save your progress, then double-click the main circuit on the explorer pane. These pins represent the inputs and outputs for adding two 4-bit numbers:

$$A_3 A_2 A_1 A_0 + B_3 B_2 B_1 B_0 = Q_3 Q_2 Q_1 Q_0$$

$A_0$  and  $B_0$  are the least significant bits of the two four-bit numbers to be added.  $Q_0$  is the least significant bit of the 4-bit sum. Use the Half\_adder and Full\_adder circuits you designed in parts 2 and 3 to implement a ripple-carry adder to perform addition between these two 4-bit numbers. T

#### **What to Submit**

Test your circuit with various inputs to make sure that addition works properly. Submit your completed HW2\_LASTNAME.circ file on Blackboard.